

1. Initialise Sample from prior to prepare initial training dataset for the NRE r_0^* : $\{\theta_n\}_0 \sim \pi(\theta)$

2. Simulate

Generate data using a simulator: $\{\theta_n\}_i \rightarrow \text{Sim}(\theta) \rightarrow \{D_n\}_i$. Create disjoint distribution $\pi \times Z$ by shuffling the pairs in joints $(\theta_n, D_n) \rightarrow (\theta_j, D_k)$ where $j \neq k$ and concatenate with joints to generate a training dataset $\{(\theta, D)\}_{i,\text{train}}$.

5. Terminate if $\mathcal{D}_{\text{KL}}$ has converged

NSNRE

3. NRE

Concatenate the newly simulated training dataset $\{(\theta, D)\}_{i,\text{train}}$ with the previous cumulative training dataset $\{(\theta, D)\}_{0:i-1,\text{train}}$. Train Neural ratio estimator $\log r_i^*$ with weights initialised from the previous run to resume training.

4. NS

Run nested sampling on $\log r_i^*(\theta, D_{\text{obs}})$ to generate $\{\theta_n\}_i$ dead samples between prior and posterior targeted at D_{obs} .